

AWS Production Setup & Deployment Guide

IAM + S3 + CloudFront + EC2 + GoDaddy + Docker/Nginx

বাংলা সংস্করণ

```
GoDaddy DNS
|
+--> bookkeepersmarketplace.store -----> Frontend :3000
+--> admin.bookkeepersmarketplace.store -> Admin :3001
+--> api.bookkeepersmarketplace.store ---> Backend :5000
      |
      Elastic IP
      |
      AWS EC2 + Nginx
      |
      MongoDB / Stripe / S3 + CloudFront
```

Reusable reference based on the Bookkeepers Marketplace deployment

দ্রুত রেফারেন্স - বর্তমান প্রজেক্ট

নিরাপত্তা: এই ডকুমেন্টে কোনো real password, secret access key বা Stripe secret রাখা হয়নি। Secrets সবসময় .env / secret manager-এ রাখবে এবং Git-এ commit করবে না।

আইটেম	বর্তমান মান
Root domain	bookkeepersmarketplace.store
Website	www.bookkeepersmarketplace.store + bookkeepersmarketplace.store
Admin	admin.bookkeepersmarketplace.store
Backend	api.bookkeepersmarketplace.store
AWS Region	us-east-2 (Ohio)
EC2 name	bookkeepers-marketplace-production
EC2 type	t3.small - 2 vCPU, 2 GB RAM
OS	Ubuntu Server 26.04 LTS, x86_64
EBS	30 GB gp3
Elastic IP	3.151.181.212
S3 bucket	bookkeepers-marketplace-prod-storage
CloudFront	d26wfj7v7cajxb.cloudfront.net

এক লাইনের পূর্ণ Flow

AWS Account -> Root MFA -> IAM User -> S3 -> CloudFront OAC -> EC2 -> Elastic IP -> GoDaddy DNS -> SSH -> Docker/Nginx -> Deploy -> SSL -> Stripe Webhook

1. AWS Account, Billing এবং Root Security

1. Client/company email দিয়ে AWS account তৈরি করো। Account ownership client-এর থাকবে।
2. Billing/Payment method client নিজে add করবে। Card number, CVV বা bank credentials developer নেবে না।
3. Root account-এ MFA enable করো।
4. Daily development/deployment-এর জন্য root login ব্যবহার করো না। আলাদা IAM user বা IAM Identity Center access ব্যবহার করো।

Rule: Root password, root MFA code, root access key এবং billing credentials কখনো client-এর কাছ থেকে নেবে না।

2. IAM Developer User তৈরি

5. AWS Console -> IAM -> Users -> Create user.
6. Username: saurav-developer (অথবা project-specific developer name).
7. AWS Management Console access enable করো।
8. Autogenerated temporary password ব্যবহার করো এবং first login-এ password change require করো।
9. Initial infrastructure setup-এর জন্য admin group / AdministratorAccess দেওয়া যেতে পারে। Setup শেষে least-privilege permission-এ নামিয়ে আনো।
10. Create user হওয়ার পর IAM sign-in URL, username এবং temporary password secureভাবে developer-কে দাও।

IAM user login-এর জন্য যা লাগে

IAM sign-in URL: `https://ACCOUNT-ID.signin.aws.amazon.com/console`
 Username: saurav-developer
 Temporary password: *****

Recommended: Human user-এর জন্য IAM Identity Center আরও ভালো long-term option। কিন্তু simple/small client setup-এ আলাদা IAM user ব্যবহার করা যায়।

3. S3 - Private Image/Video/Document Storage

11. AWS Console -> S3 -> Create bucket.
12. Bucket type: General purpose.
13. Bucket name globally unique রাখো। Current example: bookkeepers-marketplace-prod-storage.
14. Region EC2-এর সাথে একই রাখো। Current project: us-east-2 (Ohio).
15. Object Ownership: ACLs disabled / Bucket owner enforced.
16. Block Public Access: সব ON রাখো।
17. Default encryption ON রাখো (SSE-S3 শুরুতে যথেষ্ট).
18. প্রয়োজনে uploads/, documents/, videos/ prefix/folder ব্যবহার করো।

Expected behavior: Private S3 object-এর direct S3 URL browser-এ খুললে AccessDenied আসা স্বাভাবিক। Public delivery-এর জন্য CloudFront ব্যবহার করো।

4. S3 Access Key কোথা থেকে পাবে

19. IAM -> Users -> developer/s3-uploader user -> Security credentials.
20. Access keys -> Create access key.
21. Application running outside AWS / suitable use case select করো.
22. Create access key চাপলে Access key ID এবং Secret access key পাওয়া যাবে।

23. Secret access key শুধু creation-এর সময় একবার full দেখা যায় - তখনই secureভাবে save করো।

```
AWS_ACCESS_KEY_ID=AKIA...
AWS_SECRET_ACCESS_KEY=...
AWS_BUCKET_NAME=bookkeepers-marketplace-prod-storage
AWS_REGION=us-east-2
AWS_PUBLIC_BASE_URL=https://d26wfj7v7cajxb.cloudfront.net
```

Production best practice: Backend যদি EC2-তে চলে, long-lived access key-এর বদলে EC2 IAM Role দিয়ে S3 access দেওয়া বেশি secure। তখন AWS_ACCESS_KEY_ID/AWS_SECRET_ACCESS_KEY server .env-এ লাগবে না।

5. CloudFront + OAC - Private S3 public delivery

24. CloudFront -> Create distribution.
25. Available plan থেকে project অনুযায়ী plan select করো (pricing সময়ের সাথে বদলাতে পারে).
26. Distribution name: bookkeepers-marketplace-cdn.
27. Origin type: Amazon S3.
28. S3 origin: bookkeepers-marketplace-prod-storage.
29. Origin path blank রাখো - পুরো bucket origin হবে.
30. Allow private S3 bucket access to CloudFront enable করো.
31. Origin Access Control (OAC) ব্যবহার করো এবং Sign requests enable রাখো.
32. AWS-কে bucket policy update করতে দাও / generated policy apply করো.
33. Viewer protocol HTTPS / redirect HTTP to HTTPS রাখো.
34. Distribution deploy হলে CloudFront domain copy করো.

```
AWS_PUBLIC_BASE_URL=https://d26wfj7v7cajxb.cloudfront.net
```

```
Object key: uploads/profile.jpg
Public URL: https://d26wfj7v7cajxb.cloudfront.net/uploads/profile.jpg
```

Database tip: Database-এ full S3 URL না রেখে object key যেমন uploads/profile.jpg save করলে পরে CloudFront/custom domain বদলানো সহজ হয়।

6. EC2 Server তৈরি

Setting	Recommended / Current
Region	us-east-2 (Ohio) - S3-এর সাথে same region
Name	bookkeepers-marketplace-production
OS	Current: Ubuntu 26.04 LTS x86_64; future conservative default: Ubuntu 24.04 LTS if dependencies are not yet

Setting	Recommended / Current
	tested
Instance	t3.small - 2 vCPU, 2 GB RAM
Storage	30 GB gp3
Key pair	bookkeepers-marketplace-key, RSA, .pem
SSH	TCP 22 -> My IP
HTTP	TCP 80 -> Anywhere IPv4
HTTPS	TCP 443 -> Anywhere IPv4

35. EC2 -> Instances -> Launch instance.

36. উপরের settings অনুযায়ী instance configure করো।

37. 3000/3001/5000/27017/6379 public Security Group-এ open করো না। Nginx/internal networking ব্যবহার করো।

38. Launch instance -> Running -> status checks passed হওয়া পর্যন্ত অপেক্ষা করো।

Changing size later: t3.small থেকে t3.medium/large পরে change করা যায়: Stop instance -> Change instance type -> Start. Storage বাড়ানো যায়; কমানো সহজ নয়।

7. Elastic IP - Static Server IP

39. EC2 -> Network & Security -> Elastic IP addresses.

40. Allocate Elastic IP address.

41. Actions -> Associate Elastic IP address.

42. Resource type: Instance.

43. Instance: bookkeepers-marketplace-production.

44. Private IP auto-selected value রাখো এবং Associate করো।

Current project Elastic IP: 3.151.181.212

Important: Elastic IP allocate করলেই হবে না - instance-এর সাথে Associate করতে হবে। Associate না করলে SSH timeout এবং DNS pointing ভুল হতে পারে।

8. GoDaddy DNS Setup

Type	Name	Value
A	@	3.151.181.212

Type	Name	Value
A	admin	3.151.181.212
A	api	3.151.181.212
CNAME	www	bookkeepersmarketplace.store

- NS, SOA, _domainconnect, _dmarc records অকারণে change করবে না।
- Root domain -> Frontend, admin subdomain -> Admin Dashboard, api subdomain -> Backend API.
- DNS verification: nslookup bookkeepersmarketplace.store / admin... / api...

9. Windows থেকে SSH Login

Ubuntu EC2 login: Root password লাগে না। Default username সাধারণত ubuntu এবং authentication হয় .pem key দিয়ে।

```
ssh -i ".\bookkeepers-marketplace-key.pem" ubuntu@3.151.181.212
```

Windows .pem permission error fix - CMD

```
icacls ".\bookkeepers-marketplace-key.pem" /inheritance:r
icacls ".\bookkeepers-marketplace-key.pem" /remove "NT AUTHORITY\Authenticated Users"
icacls ".\bookkeepers-marketplace-key.pem" /remove "BUILTIN\Users"
whoami
icacls ".\bookkeepers-marketplace-key.pem" /grant:r "%USERDOMAIN%\%USERNAME%:(R)"
icacls ".\bookkeepers-marketplace-key.pem"
```

PowerShell variant

```
icacls ".\bookkeepers-marketplace-key.pem" /inheritance:r
icacls ".\bookkeepers-marketplace-key.pem" /remove "NT AUTHORITY\Authenticated Users"
icacls ".\bookkeepers-marketplace-key.pem" /remove "BUILTIN\Users"
icacls ".\bookkeepers-marketplace-key.pem" /grant:r "${env:USERDOMAIN}\${env:USERNAME}:(R)"
```

If location/computer changes: SSH source My IP হলে অন্য computer/country থেকে login করার আগে Security Group inbound SSH rule-এর Source আবার My IP করে update করো।

10. Ubuntu Server Basic Setup

```
sudo apt update
sudo apt upgrade -y
```

```
sudo apt install -y git curl unzip nginx
sudo systemctl enable nginx
sudo systemctl start nginx
```

Docker + Compose

```
sudo apt install -y docker.io docker-compose-v2
sudo systemctl enable docker
sudo systemctl start docker
sudo usermod -aG docker ubuntu
exit
```

- Logout করার পর আবার SSH login করো যাতে docker group permission apply হয়।
- Check: docker --version, docker compose version, docker ps.

2 GB Swap (t3.small-এর জন্য useful)

```
sudo fallocate -l 2G /swapfile
sudo chmod 600 /swapfile
sudo mkswap /swapfile
sudo swapon /swapfile
echo "/swapfile none swap sw 0 0" | sudo tee -a /etc/fstab
free -h
```

11. Frontend + Admin + Backend Deployment Structure

```
/home/ubuntu/bookkeepers-marketplace/
├── frontend/      # Next.js -> 3000
├── admin/         # Next.js -> host 3001
├── backend/       # NestJS -> 5000
└── docker-compose.yml
```

45. mkdir -p ~/bookkeepers-marketplace && cd ~/bookkeepers-marketplace
46. GitHub private repos হলে Deploy Key অথবা secure GitHub authentication configure করো.
47. Frontend, Admin এবং Backend repo clone করো.
48. .env / .env.production আলাদা করে configure করো.
49. Docker Compose-এ host ports 127.0.0.1-এ bind করো যাতে app ports public না হয়.

```
services:
  frontend:
    build: ./frontend
    restart: unless-stopped
    ports:
      - "127.0.0.1:3000:3000"

  admin:
    build: ./admin
    restart: unless-stopped
    ports:
      - "127.0.0.1:3001:3000"

  backend:
    build: ./backend
```

```
restart: unless-stopped
ports:
  - "127.0.0.1:5000:5000"
```

12. Environment Variables - Safe Template

```
# Backend
NODE_ENV=production
PORT=5000
MONGODB_URI=...
JWT_SECRET=...

AWS_BUCKET_NAME=bookkeepers-marketplace-prod-storage
AWS_REGION=us-east-2
AWS_PUBLIC_BASE_URL=https://d26wfj7v7cajxb.cloudfront.net
# Only if your architecture still uses IAM access keys:
AWS_ACCESS_KEY_ID=...
AWS_SECRET_ACCESS_KEY=...

STRIPE_SECRET_KEY=sk_test_...
STRIPE_WEBHOOK_SECRET=whsec_...

FRONTEND_URL=https://bookkeepersmarketplace.store
ADMIN_URL=https://admin.bookkeepersmarketplace.store
```

```
# Frontend
NEXT_PUBLIC_API_URL=https://api.bookkeepersmarketplace.store
NEXT_PUBLIC_STRIPE_PUBLISHABLE_KEY=pk_test_...

# Admin
NEXT_PUBLIC_API_URL=https://api.bookkeepersmarketplace.store
NEXTAUTH_URL=https://admin.bookkeepersmarketplace.store
NEXTAUTH_SECRET=...
```

Never commit: .env, .env.production, *.pem, AWS secret key, Stripe secret key, database password.

13. Nginx Domain Routing

Domain	Internal app
bookkeepersmarketplace.store + www	127.0.0.1:3000
admin.bookkeepersmarketplace.store	127.0.0.1:3001
api.bookkeepersmarketplace.store	127.0.0.1:5000

```
server {
    listen 80;
    server_name api.bookkeepersmarketplace.store;
```

```

client_max_body_size 50M;

location / {
    proxy_pass http://127.0.0.1:5000;
    proxy_http_version 1.1;
    proxy_set_header Host $host;
    proxy_set_header X-Real-IP $remote_addr;
    proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;
    proxy_set_header X-Forwarded-Proto $scheme;
}
}

```

- Frontend এবং Admin-এর জন্য একই pattern, শুধু server_name এবং proxy_pass port পরিবর্তন করো।
- Enable site -> nginx -t -> reload nginx.

```

sudo nginx -t
sudo systemctl reload nginx

```

14. SSL / HTTPS

```

sudo apt install -y certbot python3-certbot-nginx

sudo certbot --nginx -d bookkeepersmarketplace.store -d www.bookkeepersmarketplace.store
sudo certbot --nginx -d admin.bookkeepersmarketplace.store
sudo certbot --nginx -d api.bookkeepersmarketplace.store

sudo certbot renew --dry-run

```

Before Certbot: DNS records অবশ্যই EC2 Elastic IP-তে resolve করতে হবে এবং Security Group-এ 80/443 open থাকতে হবে।

15. MongoDB, S3 এবং Stripe Final Connection

- **MongoDB Atlas:** Network Access-এ EC2 Elastic IP /32 allow করো; সম্ভব হলে 0.0.0.0/0 avoid করো।
- **S3:** Private bucket রাখো; app upload করবে S3-তে এবং read URL CloudFront base URL দিয়ে তৈরি করবে।
- **Stripe:** Test mode-এ integration complete করো। Backend live হওয়ার পর webhook endpoint যেমন `https://api.bookkeepersmarketplace.store/api/v1/stripe/webhook` configure করো।
- **Live Stripe:** Client business verification complete হলে pk_live/sk_live এবং live webhook signing secret আলাদাভাবে configure করো।

16. Common Problems এবং Quick Fix

Problem	Likely reason / fix
SSH timeout	Elastic IP instance-এর সাথে associate হয়নি, port 22 blocked, অথবা SSH Source My IP পুরনো IP.
UNPROTECTED PRIVATE KEY FILE	Windows .pem ACL বেশি open - icaccls দিয়ে

Problem	Likely reason / fix
	inheritance/users remove করে নিজের user-কে Read দাও.
Permission denied (publickey)	Wrong key, wrong username, অথবা key file read permission সমস্যা. Ubuntu AMI হলে username সাধারণত ubuntu.
S3 direct URL AccessDenied	Bucket private - expected. CloudFront OAC URL ব্যবহার করো.
CloudFront NXDOMAIN	Distribution এখনও deploying অথবা domain typo. Deploy complete হওয়া পর্যন্ত wait, domain copy করে test, প্রয়োজনে ipconfig /flushdns.
Domain opens but app fails	Nginx config, Docker container status, internal port, .env এবং application logs check করো.
Other country/PC থেকে SSH হয় না	Security Group SSH Source নতুন My IP দিয়ে update করো.

17. Final Production Checklist

- Root MFA enabled; root credentials developer-এর কাছে নেই.
- Developer IAM access separate; setup শেষে permissions reduce করা হয়েছে.
- S3 private; Block Public Access ON; CloudFront OAC configured.
- EC2 Security Group: 22 My IP, 80/443 public; app/database/Redis ports public নয়.
- Elastic IP associated এবং GoDaddy DNS correct.
- Docker containers restart policy configured.
- Nginx routing works for root/admin/api domains.
- HTTPS certificates installed and renewal test passed.
- MongoDB network restriction configured.
- Secrets Git-এ নেই; .pem safe location-এ.
- AWS Budget/Cost alert configure করা হয়েছে.
- Unused Elastic IP / EC2 / storage resources remove করা হয়েছে.

Cost reminder: EC2 instance, EBS storage, public IPv4/Elastic IP, S3 storage/requests এবং CloudFront usage অনুযায়ী charge হতে পারে। Unused resources cleanup করো এবং Budget alert সেট করো।